ET AI HACKATHON 2026 · PROBLEM STATEMENT 4

PRAMAAN

EPC Deviation Intelligence for Hyperscale Data Centres

Spec-to-Site Deviation Sentinel + Commissioning Risk Twin

*Pramaan (प्रमाण) — Sanskrit for **proof**. Named on purpose: every number here is reproducible.*

50	1,024	1.000	12
Deviations Caught	Weeks Saved	Real-doc Recall	Projects

THE PROBLEM

Deviations Hide in Thousands of Pages

In a **40 MW Tier IV** data centre build, the design basis, vendor submittals, and governing standards live in different systems, reviewed by different people.

Today	With Pramaan
Week 16–44	**Week 11**
Deviations surface during commissioning — rework, schedule delays, cost overruns	All 14 caught the day the submittal was uploaded — under 5 minutes

WHAT GOES WRONG

14 Critical Deviations in One Project

DEVIATION	SPEC SAYS	VENDOR SUBMITTED	IMPACT	LEAD
UPS battery runtime	10 min	7 min	Tier IV fault tolerance broken	27w
Generator fuel autonomy	24 h	12 h	Design-duration outage failure	30w
Cooling redundancy	N+2	N+1	No concurrent maint. tolerance	28w
BMS monitoring	Dual	Single	Single point of failure	33w
Generator start time	10 s	15 s	Battery depletion risk	23w
BMS alarm coverage	Complete	Missing leak	IST-14 alarm test will fail	29w
Cable fire rating	CMP	CMR	NFPA 75 violation in plenum	11w

Showing 7 of 14 · Total lead time: **267 weeks** (5+ years)

THE HEADLINE

267 weeks

of total lead time saved across 14 deviations

> *Pramaan caught a critical BMS monitoring single-point-of-failure **33 weeks** before it would have failed the full-facility failover drill. That's the difference between a one-line email and a seven-figure schedule slip.*

THE SOLUTION

5 AI Agents, One Pipeline

LangGraph orchestrator cross-references every requirement against every submittal against every governing standard

AGENT 1	→	AGENT 2	→	AGENT 3 — THE BRAIN	→	AGENT 4	→	AGENT 5
Ingestion		**Extraction**		**Reconciliation**		**Cx Predictor**		**RFI Copilot**
PDF → Markdown		Structured Triples		Cross-doc reasoning		Test + Lead Time		RAG Q&A + Citations

LangGraph StateGraph · Conditional routing · TypedDict state · Validation gate

DETECTION ACCURACY

Honest Detection Numbers

The synthetic portfolio below scores 1.000 *by construction* — a plumbing / breadth check, not a flex.

1.000	1.000	1.000	0
Precision	Recall	F1 Score	False Positives

1.000	17	160+
Cx Prediction	Cx Tests Mapped	Automated Tests

The number that matters: 11 sourced real datasheet pairs — 19 deviations (recall 1.000), 0 false positives, and a self-scored ≈0.9 on one contested ASHRAE case (live-verified, gemini-2.5-flash). We report ~0.9, not a suspicious 1.000.

MULTI-PROJECT GENERALISATION

12 Projects · 11 Countries · 6 Tier Standards

Synthetic breadth test — deviations are seeded, so F1 is 1.000 *by construction*. The real-world proof is the sourced-datasheet pairs, not this grid.

Meghdoot
Tier IV · Mumbai · 40MW
14 devs · 267w saved
F1 = 1.000

Vajra
Tier III · Pune · 20MW
4 devs · 95w saved
F1 = 1.000

Nordic Edge
EN 50600 · Oslo · 10MW
5 devs · 113w saved
F1 = 1.000

Sahara
Tier II · Dubai · 5MW
3 devs · 38w saved
F1 = 1.000

Cascade
Tier IV · Oregon · 30MW
4 devs · 80w saved
F1 = 1.000

Yangtze
GB 50174 · Shanghai · 50MW
3 devs · 98w saved
F1 = 1.000

Athena
EN 50600 · Frankfurt · 15MW
4 devs · 80w saved
F1 = 1.000

Sakura
JEITA · Tokyo · 25MW
3 devs · 58w saved
F1 = 1.000

Outback
Tier III · Sydney · 8MW
3 devs · 53w saved
F1 = 1.000

Maple
Tier III · Toronto · 12MW
3 devs · 58w saved
F1 = 1.000

Pampas
Tier II · São Paulo · 6MW
2 devs · 34w saved
F1 = 1.000

Thames
Tier IV · London · 35MW
2 devs · 50w saved
F1 = 1.000

1,024
Total Weeks Saved

256
Total MW

25+
Standards Covered

LIVE DASHBOARD

22-Section Dashboard · 27 Components

AWARENESS	INTELLIGENCE	ACTION
Hero Intro & Problem	System Health Grid	Live PDF Analysis
Sentinel Alert	Deviation Register	RFI Copilot (RAG)
267-Week Counter	Risk Matrix	ROI Calculator
Before / After Toggle	Document Diff Viewer	Export Evidence Pack
	Cx Risk Twin (Gantt)	Multi-Project Eval

Next.js 15 · Dark theme · Scroll-reveal animations · Responsive · Streaming AI

EVAL HARNESS

Reproducibility checks + proof on real documents

Paths 1–2 are seeded-corpus integrity checks (1.000 *by construction*, not detection skill); Path 3 — real, unseen datasheets — is the credibility signal.

Path 1: Structured

Pre-extracted triples vs ground truth

F1 = 1.000

Data integrity proof

Path 2: Text-Based

Raw markdown → regex extraction

F1 = 1.000

Independent input path

Path 3: LLM Agent

Real, unseen datasheets → LLM reasoning → deviations

Recall 1.000

19/19 hard · 0 FP · ~0.9 precision (live-verified)

`make eval-all` — two reproducibility checks + the real-document proof

DON'T TRUST US — RUN IT

Every Number on These Slides Reproduces Offline

One command. No API key. A judge's laptop is enough.

`git clone · make verify`

- **446 tests** — API, agents, security, resilience, real pairs
- **3 eval harnesses** — every headline metric regenerated
- **Type-check + production build** — CI green, publicly visible
- **No-key real-datasheet harness** — `eval/real_pairs_offline.py`

We surveyed ~70 public repos across every track of this hackathon: **zero CI pipelines, zero eval harnesses, one minimal test suite.**
Claims are easy. Pramaan means proof.

SCALE STORY

Demo → Enterprise

Current Demo

- 12 projects, 11 countries
- 81 systems across portfolio
- 203 requirements tracked
- 50 deviations detected
- TF-IDF retriever
- Synchronous agents

At Enterprise Scale

- Hundreds of concurrent projects
- 500+ systems per project
- 14,000+ line items per project
- Continuous monitoring pipeline
- pgvector / Qdrant vector store
- LangGraph async + task queue

HACKATHON RUBRIC

How Pramaan Maps to Every Dimension

Innovation

Beyond AI submittal review: predicts which commissioning test each deviation fails and how many weeks early. Cross-references spec + submittal + standard with a citation chain. 5 agents, LangGraph orchestration.

Business Impact

1,024 weeks of early detection across 50 findings in 12 projects. Interactive ROI calculator shows ₹1,788 lakhs avoided on ₹800 Cr project.

Technical Excellence

Synthetic portfolio is 1.000 by construction (breadth check); the real signal is 11 sourced datasheet pairs — 19 deviations (recall 1.000), 0 false positives, self-scored ≈0.9 on one contested case. 435 tests. 28 endpoints. Independent text + real-LLM evaluation.

Scalability & UX

22-section dashboard with 60-second demo narrative. Multi-project eval. Streaming AI. PDF upload. Evidence export. Responsive design.

TECH STACK

Built With

LLM Gemini 2.5 Flash	**ORCHESTRATION** LangGraph	**BACKEND** FastAPI + SSE
FRONTEND Next.js 15 + React 19	**PDF** pdfplumber + PyMuPDF	**RETRIEVAL** TF-IDF → pgvector
EVAL Triple Harness	**SCRAPING** Firecrawl + Playwright	**DEPLOY** Vercel + Render

BUSINESS IMPACT

The ROI That Sells Itself

₹800 Cr	**₹1,788L**	**< 30**
Project Value	Rework Avoided	Payback (Days)

Every deviation carries `lead_time_weeks`.
The metric that wins is **time** — not accuracy alone.

THANK YOU

PRAMAAN

50 deviations · 1,024 weeks saved · 12 projects · 11 countries

Real-doc: 0 false positives · self-scored ~0.9 · 446 tests

▶ Live Demo

API Docs

parth-tan.vercel.app · parth-3puc.onrender.com